

NEXAFORGE AI

Grounded RAG Quality Engineering Playbook

End-to-end practices for ingestion, retrieval, grounding, citation, evaluation, and continuous improvement

Document ID	Version	Owner	Classification
NF-AI-003	2.0	Knowledge Systems Engineering	Confidential - Internal Use

Purpose

Provide a practical engineering standard for retrieval-augmented generation systems that answer questions from enterprise documents. The playbook covers corpus preparation, indexing, hybrid retrieval, evidence selection, answer generation, citation integrity, evaluation, and production monitoring.

Keywords: RAG, retrieval, grounding, citations, chunking, reranking, evaluation

Document Control

Field	Value
Status	Approved methodology baseline
Effective date	21 July 2026
Review cycle	Every 12 months or upon material change
Primary owner	Knowledge Systems Engineering
Related standards	Security, privacy, software delivery, incident response, and data governance standards
Supersedes	Version 1.0 concise edition

How to Use This Document

Teams should use this document as a design and review guide, not as a substitute for engineering judgment. Sections describe mandatory controls, recommended practices, decision points, and evidence expected at release. Tailoring is permitted when documented and approved by the accountable owner. Any deviation that increases risk must include compensating controls and a review date.

1. Purpose and Objectives

Provide a practical engineering standard for retrieval-augmented generation systems that answer questions from enterprise documents. The playbook covers corpus preparation, indexing, hybrid retrieval, evidence selection, answer generation, citation integrity, evaluation, and production monitoring.

2. Scope and Applicability

This standard applies to new implementations, major changes, vendor replacements, and material expansions of systems that rely on grounded rag quality engineering playbook. It covers prototypes that process real organizational data, pre-production pilots, and production services. Small experiments using synthetic data may follow a reduced process, but they must still document ownership, purpose, and data handling. The methodology does not replace legal, security, privacy, or domain-specific requirements; instead, it provides the engineering structure through which those requirements are implemented and verified.

3. Core Principles

Retrieval is a product capability: Quality depends on document coverage, metadata, freshness, and permissions as much as model choice.

No evidence, no claim: Material claims must be supported by retrieved evidence or explicitly labeled as general knowledge.

Optimize the pipeline, not one metric: Recall, precision, answer faithfulness, latency, and cost must be balanced together.

Respect document structure: Headings, tables, lists, and version metadata are essential signals, not formatting noise.

Evaluate realistic questions: Test sets should reflect actual user language, ambiguity, and multi-document synthesis.

4. Lifecycle and Detailed Procedure

1. Corpus qualification

Inventory sources, ownership, update frequency, access rules, document formats, duplication, and expected question types. Exclude obsolete or legally restricted materials before indexing.

- Required evidence: a reviewable artifact or trace showing that corpus qualification was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

2. Parsing and normalization

Extract text, tables, headings, page numbers, links, and document metadata. Normalize encoding and whitespace while preserving semantic structure. Record parser confidence and failures.

- Required evidence: a reviewable artifact or trace showing that parsing and normalization was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

3. Chunk design

Choose chunk boundaries based on document structure and answer granularity. Use overlap only when necessary. Attach document title, section path, page, version, dates, and access labels to every chunk.

- Required evidence: a reviewable artifact or trace showing that chunk design was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

4. Index construction

Create dense embeddings and a lexical index. Configure language analyzers, stop words, stemming, vector dimensions, and metadata filters. Version the index build and embedding model.

- Required evidence: a reviewable artifact or trace showing that index construction was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

5. Query understanding

Rewrite only when it improves retrieval. Extract entities, dates, product names, and access context. Generate subqueries for multi-part or comparative questions while preserving the original intent.

- Required evidence: a reviewable artifact or trace showing that query understanding was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

6. Candidate retrieval and reranking

Retrieve broad candidates with hybrid search, then apply metadata filters, deduplication, diversity controls, and a reranker. Keep enough evidence for multi-document synthesis but avoid flooding the generator.

- Required evidence: a reviewable artifact or trace showing that candidate retrieval and reranking was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

7. Grounded answer generation

Provide the model with a strict answer contract: use only supplied evidence for corpus-specific claims, identify uncertainty, avoid unsupported extrapolation, and cite each material claim.

- Required evidence: a reviewable artifact or trace showing that grounded answer generation was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

8. Citation validation

Check that cited passages actually entail the associated claims, that page references exist, and that no citation points to an inaccessible or outdated source.

- Required evidence: a reviewable artifact or trace showing that citation validation was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

9. Evaluation and monitoring

Measure retrieval and generation separately, then run end-to-end tests. Monitor no-answer rate, citation failures, source drift, parser errors, and query clusters with poor outcomes.

- Required evidence: a reviewable artifact or trace showing that evaluation and monitoring was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

5. Entry Criteria

Work may enter the methodology when a named owner, defined user outcome, initial data classification, and target environment are available. Teams should also identify downstream consumers, irreversible actions, expected traffic, and the cost of wrong answers. If these inputs are unknown, the first deliverable is a discovery brief rather than an implementation.

6. Design Review Expectations

The design review should focus on concrete decisions and evidence. Reviewers inspect architecture diagrams, state or data schemas, model and tool dependencies, evaluation plans, privacy boundaries, fallback behavior, and operating metrics. Open issues must have owners and dates. A design is not approved merely because a model can demonstrate the happy path; the team must show how the system behaves with missing inputs, conflicting evidence, unavailable services, malformed outputs, and unauthorized requests.

7. Documentation and Traceability

Each release maintains a versioned record of requirements, decisions, prompts or policies, model identifiers, data or index versions, test results, known limitations, and approvers. Traceability should allow an incident responder to determine which configuration produced a specific output. Documentation must be concise enough to maintain but detailed enough for a new engineer to reproduce critical behavior.

8. Change Management

Changes are categorized as low, medium, or high impact. Low-impact changes include wording or monitoring adjustments with no behavior change. Medium-impact changes affect routing, prompts, thresholds, or non-sensitive data. High-impact changes include model replacement, new write capabilities, new sensitive data, changed jurisdictions, or altered decision authority. Medium and high-impact changes require targeted regression testing; high-impact changes repeat the relevant risk and approval reviews.

9. Operational Readiness

Before launch, the team confirms monitoring, alert ownership, on-call procedures, rollback, rate limits, access controls, data retention, incident communication, and user support. Synthetic probes should continuously test critical paths. Runbooks should describe both technical recovery and the user-facing behavior during degradation.

10. Continuous Improvement

Production evidence is reviewed on a regular cadence. Teams analyze failures by root cause rather than by surface symptom, distinguish model errors from retrieval, tool, data, policy, or interface errors, and prioritize improvements by user harm and frequency. Confirmed defects become regression tests. Metrics are segmented by use case, language, tenant, and risk tier to avoid hiding poor performance inside global averages.

12. Roles and Responsibilities

Role	Accountability
Corpus owner	Approves source inclusion and lifecycle.
Ingestion engineer	Builds parsing, normalization, and indexing.
Retrieval engineer	Tunes search, filters, and reranking.
LLM engineer	Defines grounded generation and citation behavior.
Domain expert	Creates gold answers and relevance judgments.
Security engineer	Enforces document-level permissions.

13. Measurement Framework

Metric	Definition and Use
Recall@k	Relevant evidence appearing in the top k results.
nDCG@k	Rank quality with graded relevance.
Context precision	Share of supplied context that is useful for the answer.
Answer faithfulness	Claims entailed by cited context.
Citation correctness	Citations that support the nearby claim.
Answer completeness	Required answer elements covered.
Freshness compliance	Answers using the latest valid source when versions exist.
Permission violation rate	Results exposed outside authorized scope.

Metrics must be segmented by use case and risk tier. Teams should define a baseline, target, alert threshold, and owner for each production metric. A metric without an action rule is considered observational rather than operational.

14. Worked Example

An employee asks whether a vendor can store customer data outside the EU. The query planner identifies vendor name, geography, and policy intent. Hybrid retrieval finds the data residency policy, vendor addendum, and a newer exception memo. The reranker places the exception memo first because it specifically names the vendor. The answer explains the general rule, the approved exception, its expiration date, and cites all three documents. A freshness check prevents an older policy from overriding the newer memo.

15. Risk Register and Controls

Risk	Primary Controls
Missing documents	Maintain coverage dashboards and source-owner attestations.
Bad parsing	Store parse diagnostics and route complex PDFs or tables through specialized extraction.
Chunk fragmentation	Use structure-aware chunks and parent-child retrieval.
Embedding blind spots	Combine dense and lexical retrieval, especially for IDs and exact terms.
Citation theater	Validate entailment rather than merely attaching references.
Access leakage	Apply authorization before retrieval and again before response assembly.

16. Release Gate

- A named business and technical owner has approved the intended use and operating boundaries.
- Required architecture, security, privacy, and risk reviews are complete.
- Evaluation results meet minimum thresholds and no severe unresolved defect remains.
- Monitoring, alerting, rollback, and incident procedures have been tested.
- User-facing disclosures, approval checkpoints, and fallback behavior are implemented where required.
- All model, prompt, tool, data, index, and policy versions are recorded in the release manifest.

17. Review Checklist

- Is the user outcome specific, measurable, and appropriate for AI assistance?
- Are authority boundaries and prohibited actions explicit?
- Can every important output be traced to inputs, evidence, and configuration?
- Does the design handle missing, conflicting, stale, or malicious inputs?
- Are sensitive data, permissions, retention, and deletion controlled?
- Are severe failures represented in the evaluation suite?
- Can the service degrade or stop safely when dependencies fail?
- Are metrics connected to owners and response procedures?

18. Appendices

Chunk metadata schema

This appendix should be maintained as a project-specific artifact. It records the concrete implementation details, decisions, evidence, and approvals needed to apply the Grounded RAG Quality Engineering Playbook in a reproducible manner.

- Owner and review date
- Inputs and dependencies
- Decision criteria
- Required evidence
- Known limitations and follow-up actions

RAG test question taxonomy

This appendix should be maintained as a project-specific artifact. It records the concrete implementation details, decisions, evidence, and approvals needed to apply the Grounded RAG Quality Engineering Playbook in a reproducible manner.

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- Inputs and dependencies
- Decision criteria
- Required evidence
- Known limitations and follow-up actions

Citation validation checklist

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- Inputs and dependencies
- Decision criteria
- Required evidence
- Known limitations and follow-up actions

Index release checklist

This appendix should be maintained as a project-specific artifact. It records the concrete implementation details, decisions, evidence, and approvals needed to apply the Grounded RAG Quality Engineering Playbook in a reproducible manner.

- Owner and review date
- Inputs and dependencies
- Decision criteria
- Required evidence
- Known limitations and follow-up actions

Revision History

Version	Date	Summary
1.0	21 July 2026	Initial concise methodology edition.

2.0	21 July 2026	Expanded edition with detailed lifecycle, controls, roles, metrics, examples, risks, release gates, and appendices.
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